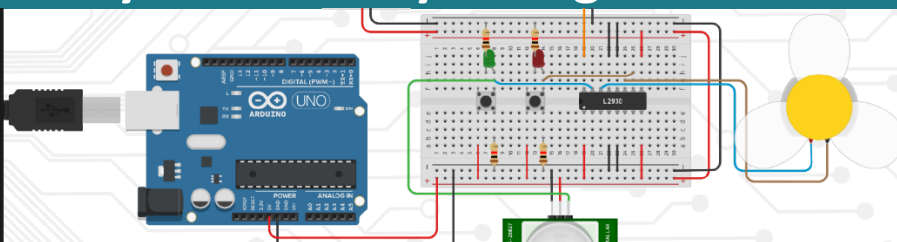


Moraine Valley Community College

LAN 120 IoT Fundamentals



Course Identification

Date:	October 19, 2026
Semester:	Fall
Course Title:	IoT Fundamentals I Course Number: LAN.120.300
Credit Hours:	3 Credit Hours Total Contact Hours: 4 (2 Lecture & 2 Lab)
Course Meets:	Online
Prerequisites:	None
Catalog Description:	The Internet of Things (IoT) is a large, connected network of physical devices that collect and exchange data via the Internet. In this course, students will learn about the Internet of Things, how it is used by a wide range of industries and its application within each. Students will also learn about IoT components including hardware, software, and other peripheral devices. Students will design, build, and program IoT devices. Relevant communication protocols and networking concepts will be covered.
Course Site:	http://canvas.morainevalley.edu (Please note: There is no “www” in the URL)
Refund Deadline:	10/22/26 Withdraw Deadline: 11/29/2026

Faculty Information

Office Hours:	Tuesday: 1:00pm – 3:00pm (T906) Thursday: 11:00 – 1:00 pm (Online)
Office Phone:	708.974.5512 (Please note: This phone number does not accept text messages)
E-Mail:	christensen@morainevalley.edu

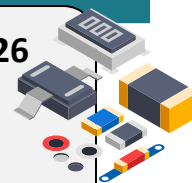
Major Course Objectives

1. IoT applications
2. IoT standards and specifications
3. IoT enabling technologies and components
4. IoT hardware, software, and electronic components
5. IoT communication protocols and networking
6. IoT privacy and security

FacultyHack@Gateways 2026

Revised Syllabus:

The revised LAN 120 syllabus adds Jupyter notebooks, Science Gateways, public datasets, cloud/HPC resources, data visualization, and AI-assisted analysis, helping students connect Arduino sensor projects to cybersecurity, critical infrastructure, and scientific computing.



Expected Outcomes for Student Learning

A. Common Learning Outcome

Information and Technology Literacy: Effectively find, evaluate, manage, transform, and exchange information using a variety of technologies.

B. General Education Learning Outcomes

Upon successful completion of this course, a student will be able to:

1. Define the Internet of Things (IoT) and explain its significance.
2. Identify IoT standards and specifications.
3. Identify and describe key enabling IoT technologies and components.
4. Identify IoT hardware and electronic components and explain how they work.
5. Understand how sensors and embedded systems work.
6. Compare and contrast various IoT hardware and software.
7. Design, build, and program an IoT device.
8. Describe IoT communication protocols and networking concepts.
9. Discuss security and privacy issues related with the implementation of IoT.

Addition: BRIDGE-CI: From Sensors to Scientific Computing (NEW)

BRIDGE-CI extends the existing LAN 120 experience from building and programming individual IoT devices to understanding what happens to sensor data after it is collected. Students will be introduced to Science Gateways, Jetstream2, Jupyter notebooks, public datasets, cloud/high-performance computing (HPC), data visualization, AI-assisted anomaly detection, cybersecurity, and critical infrastructure applications. Activities are scaffolded for students with no prior Python or HPC experience.

Additional BRIDGE-CI Learning Outcomes

10. Explain, at an introductory level, how Science Gateways, Jupyter notebooks, cloud computing, and HPC support data-intensive research and IoT applications.
11. Use a scaffolded Jupyter notebook to load, inspect, and visualize authentic sensor or IoT data.
12. Modify selected notebook variables or parameters and explain how those changes affect the resulting analysis or visualization.
13. Interpret patterns, trends, thresholds, and anomalies in sensor or cybersecurity data and identify limitations in the evidence.
14. Explain the purpose of AI-assisted anomaly detection and distinguish false positives from false negatives in an IoT or critical infrastructure context.
15. Collect a small dataset from an Arduino-based sensor, analyze and visualize the data in a Jupyter notebook, and communicate findings in a technical report.
16. Connect sensor reliability, data integrity, cybersecurity, and critical infrastructure decision-making.

Assessment of Student Learning

Typical classroom assessment techniques for this class include:

- Projects, Group Participation, Objective Tests, and Studio / Lab Performance.

BRIDGE-CI Assessments and Activities (NEW)

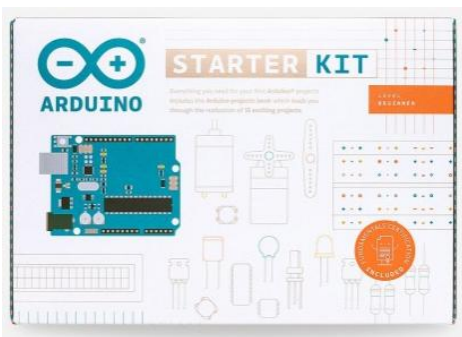
Guided Notebook 1: Sensor Data Exploration and Visualization. Students will use a prepared notebook and authentic industrial/sensor data to run cells, inspect a CSV dataset, change selected variables, create time-series visualizations, and interpret normal behavior, thresholds, and trends.

Guided Notebook 2: AI-Assisted Anomaly and Cybersecurity Detection. Students will use a prepared notebook and an IoT/cybersecurity dataset to examine model outputs, interpret a confusion matrix, and discuss false positives, false negatives, and possible operational consequences.

Revised Final Project: Student-Designed Sensor Investigation. Students will develop a research question, select and program an Arduino-compatible sensor, collect a manageable dataset, document the data in CSV format, analyze and visualize the data in a scaffolded Jupyter notebook, and submit a technical report that includes findings, limitations, and a cybersecurity or critical infrastructure connection.

Students will be evaluated on following a sound process, working with evidence, interpreting results appropriately, recognizing limitations, and communicating findings. The goal is not advanced Python programming or development of a sophisticated machine-learning model.

Required Textbook and Materials



Required Text:

Arduino Fundamentals Bundle
(Includes the Certification Exam)

\$148.21 at the bookstore

Note: You will need the **bundle** for the class – your final exam will be the **Arduino Certification Exam** which is included in this kit.

Course Policies and Procedures

Attendance Policy

You should also log into our Canvas course at least three times a week to check announcements, contribute to discussions, and complete other tasks that may be assigned and required. It is your responsibility to find out about any missed topics and assignments. Any quiz or exam not taken during its designated time will be entered as a zero into your grade record. Makeup exams are at my discretion and will be permitted only for valid, documented reasons. In all cases, they must be arranged *before* the exam is missed.

Time Commitment: Student Preparation Needed to Be Successful

To be successful in this course, the average student should spend a minimum of 1.5 to 2 hours of outside study time for every hour spent in class in addition to the time spent in class. According to that estimate, a student should spend approximately 8 hours studying for this class each week (the in-class hour equivalent is 4 hours). You may find yourself needing less time or more time, depending on your comprehension of the material. This time will be spent reading the text, completing assignments, reviewing class notes, studying for tests, etc. For guidance on the total number of credit hours to be pursued, please refer to the current MVCC catalog.

Email Accounts / Email Correspondence Policy

Students must use the email account that has been provided by Moraine Valley Community College. This is a college policy and will be enforced. More information concerning your student email account can be found using the following web address: <http://www.morainevalley.edu/studentemail>

Please note: Any correspondence with an email address other than the college issued account will not be read and will be deleted. *Caution:* If you elect to forward your MVCC email to a personal account (i.e. username@comcast.net) then do not send a response back to me using your personal account. All email should be sent directly from your MVCC account.

Please follow the policies below when sending an email (please send your email to both of us):

- Include a subject line in each email that you send. The subject line must begin with the course number and section followed by the topic. For example: LAN.120.300: Question on assignment.
- The body of the email must include at least one complete sentence and close with your name. When asking for help, please do your best to be specific about the questions that you have.
- Make sure that you read and adhere to the guidelines and procedures for using your MVCC email account. Review the policies posted at <http://www.morainevalley.edu/studentemail/>

Email Accounts / Email Correspondence Policy, cont.

Although I will try to make it a priority to respond to your e-mail messages within 24 hours, please be advised that it may take me anywhere between 36 and 48 hours to respond to your messages or voicemail during the Monday – Friday work week. Messages received after 12pm on Friday through Sunday will be answered within the average response period starting on Monday morning. Please make note of this so that any questions that you have regarding the materials or assignments can be addressed so that you can meet deadlines.

Academic Integrity

Students are expected to uphold the school's code of conduct relating to academic honesty. Students assume full responsibility for the content and integrity of the academic work they submit. The guiding principle of academic integrity shall be that a student's submitted work, examinations, and projects must be that of the student's own work. Students are expected to set high personal standards for themselves including, but not limited to, digital responsibility, and respect for the intellectual property of others.

Use of AI-Assisted and Computational Tools (NEW)

Some BRIDGE-CI activities use provided Jupyter notebooks and AI-assisted analysis tools. These outputs should be treated as evidence to examine, not as automatic truth. Students remain responsible for verifying results, explaining their own reasoning, documenting required changes to code or parameters, and submitting their own interpretations. Any use of generative AI must follow the instructions for the specific assignment and be acknowledged when required.

Cheating/Plagiarism Policy

All students are expected to demonstrate academic integrity as they complete homework assignments, exams, and final projects. The definition of academic integrity for this course requires that students submit their own work. Students should not share their work with others, allow others to complete their work for them, copy code or ideas from other unauthorized sources (i.e. websites, friends, etc.), or complete exams with unauthorized notes or assistance. For this class, it is permissible to assist classmates in general discussions of computing techniques. General advice, conversations and interactions among classmates are encouraged.

A student will be found guilty of cheating if they:

- Submit as your own the work of someone else.
- Sharing work between yourself and another classmate.
- Use or obtain unauthorized assistance in any academic work.
- Give unauthorized assistance to other students.
- Copying code from a site and turning it in as your own.

Please note: Any student caught cheating or plagiarizing may receive a grade of zero on that assignment or a failing grade for the entire course. This includes copying another student's computer file.

If you are unsure whether a situation constitutes a code of conduct violation, the student should contact the instructor to discuss it.

Student Conduct Policy

Each student is responsible for adhering to the Code of Student Conduct as stated in the college catalog. All students are expected to show honesty and integrity in the classroom, including consideration for other students and the instructor. Student policies can also be found on the college portal and our Canvas course.

General Information Sheet

The college "General Information Sheet" can be downloaded and printed from the college portal as well as our Canvas course. It is the student's responsibility to read this sheet each semester.

Student ID / Library Use / Computer Lab Use

Moraine Valley students are required to have a photo identification card in order to use certain college facilities, such as the library, Testing Center and computer labs. For more information about how to obtain a student ID, please visit the following website: <https://www.morainevalley.edu/student-success/orientation/photo-id/>

Incomplete ("I") Grades

Incompletes are provided to allow a student who experiences a **major** life event, such as emergency surgery or death of a loved one (i.e. spouse, child, or parent), to complete a course. The purpose of an incomplete is not to allow a student who is doing poorly in the course to repeat the course the following semester. Incomplete grades will only be issued with proper documentation. Please see the college catalog for the provisions for obtaining an incomplete.

Use of Electronic Devices

Students and college visitors may not use and must silence cell phones, pagers, and other communication devices in all instructional areas which include: all labs and classrooms during instructional sessions, the Learning Resource Center/Library, the Assessment Services Center and other areas designated by the college. Portable devices for listening to the radio or playing music are not to be used in the classroom, even if headphones are available.

Computer Labs

There will be weeks that you will need to work on assignments outside of class time. If you do not have access to a computer at home or at work, you can use one of the computer labs located on campus.

The following labs are available for you to use:

- T Building Open Lab in T920 (There are specific computers for the web / graphics classes that have Adobe products installed).
- CyberCafe, located in Building D
- Library/Learning Resources Center, located in Building L
- Laptop computers are available for loan through the Library. Please visit the library for more information about the laptop loan program.

If you plan on working in an open lab and want to print out any work, you will need to obtain a pay-for-print debit card. You can purchase cards with varying denominations at the LRC print card machine (located in the Library computer center). After issuing a print command in the computer lab, your print request will be placed in queue. You will then need to go to the print-release station, identify your print job by the workstation number, and swipe your print card to release the job to the printer. All jobs will be automatically deleted from the queue if they are not released after 15 minutes.

Use of MVCC computers and Internet access

Installation of personal software (ANY software not owned by the college, nor authorized by the college) is **STRICTLY PROHIBITED**. Use of the Internet is restricted to what is required for the courses being taken at the college. Any student found using MVCC computers to display or download inappropriate material will be warned once, and if this conduct continues will be dealt with according to the student conduct policy.

College Refund & Withdrawal Policy

Withdrawal Policy: A student who does not withdraw officially from a course may receive a grade of “F” depending on course progress or course attendance, which will become a part of the student’s permanent record. You are responsible for determining whether you should withdraw from the course. Please note that if you decide to withdraw from the course, you will no longer have access to the Canvas course. Lastly, please do not request an extension or an exception to this withdrawal date - it will not be granted.

Check your student email; refund and withdraw deadline dates should have been sent to you.

Institutional Drop Policy

Moraine Valley Community College has the right to institutionally drop students for non-attendance. Students who are reported at the Financial Aid Census Attendance date by their instructors as nonattending will be institutionally dropped from those courses. The course that is Institutionally Dropped will not be reflected on the student’s final transcript, and charges for the course will be reversed. Students who are receiving financial aid may have their aid impacted by courses being institutionally dropped.

Syllabus and Session Schedule Changes

The course schedule is tentative and subject to change at the instructor’s discretion. The syllabus is also subject to change. Students will be notified of changes by way of announcements in our Canvas course.

Late Assignment Policy

Assignments are to be completed and turned in as scheduled. Any assignments submitted late will have 10% deducted from the grade for each day (including weekends and holidays) after the due date. No assignments will be accepted after one week (7 days) from the specified due date, and a score of zero (0) will be recorded for that assignment.

Exam Policy

Exams are to be taken at the specified date and time. If a student cannot take the exam at the specified date and time, a makeup exam will be made available in the testing center for one (1) week, at the discretion of the instructor. If the exam is not completed during this time, a score of zero will be recorded.

Office 365 – Available to Students for Free!

As a Moraine Valley student, you are eligible to receive Microsoft Office 365 for free. The following site provides step-by-step instructions on how to install Office 365:

<https://mvcc.teamdynamix.com/TDClient/1952/Portal/KB/ArticleDet?ID=107722>

Grading Policies & Procedures

Grades are a symbol of your level of performance. Grades *are earned* by the student; grades are *not given* by the instructor. Students do not compete with one other for grades in this course. Your final grade will be based upon points earned for assignments and objective examinations. There are no points for minimum expectations, such as attendance and participation. ***There is no extra credit.***

The primary criteria used for evaluating assignments are style, structure, and accuracy. To receive full credit, an assignment must meet all the requirements outlined in the assignment. Appropriate deductions will be made from the number of points available for the assignment for any minor or major errors. You will receive feedback for each assignment that you submit.

Grade Distribution

All grading will be on a point system. Each exam and assignment will have a point value. At the end of the semester the final grade will be based on the total percentage of points earned. All students must take the final exam and turn in the final project to pass the course. This grade distribution is tentative as Arduino Challenges may be added to individual projects throughout the semester. This will be noted in the individual assignments.

The distribution of weights is as follows:

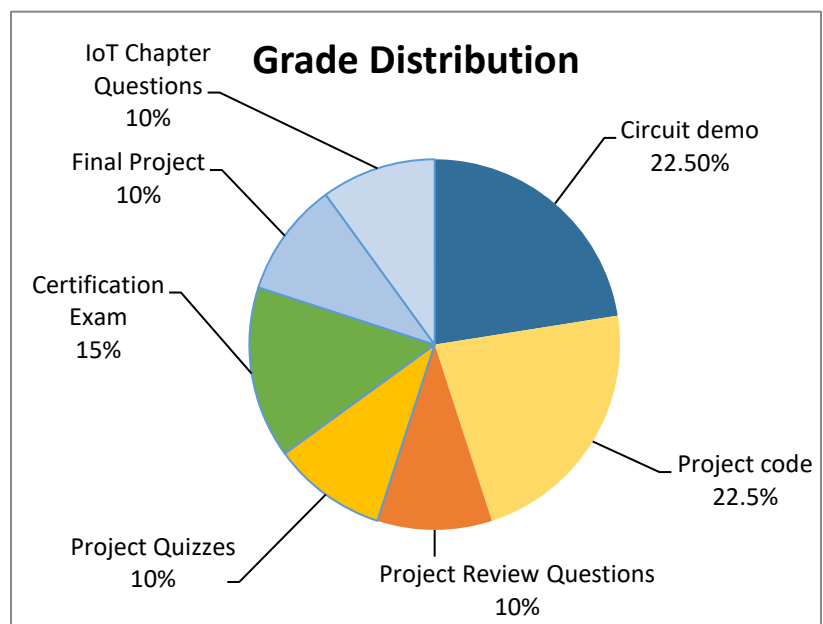
Assessment	#	Points	Weight
Circuit demo (video)	12	120 (=12 x 10 pts)	22.5%
Project Code (comments required) **	11	110 (=11 x 10 pts)	22.5%
Project Review Questions	12	60 (=12 x 5 pts)	10%
Project Quizzes	12	60 (=12 x 5 pts)	10%
IoT Chapter Questions	7	70 (=7 x 10pts)	10%
Final Project	1	100 pts	10%
Certification Exam (final exam)	1	100 pts	15%
Total		610	100%

**** to earn points for project code, you must submit a circuit demo**

Grades will be assigned as follows:

90 – 100% A
80 – 89% B
70 – 79% C
60 – 69% D
Below 60% F

The Incomplete grade (I) is reserved for students who encounter serious personal, professional, or medical problems. An Incomplete (I) is not a way out for a student who has not managed his/her time well. A request for an Incomplete must be submitted to the instructor, in writing, by the end of the twelfth week of the semester. Emergencies will be handled as necessary.



Tentative Course Schedule

Week	Topic	Chapter / Projects	Assessments
1	Syllabus Overview Introduction to Arduino: Electronics, IDE setup, Arduino board IoT Overview & Tinkercad demo Projects 1 & 2: Electronics, Tinkercad, and Coding NEW Addition BRIDGE-CI Introduction: From Sensors to Scientific Computing Science Gateways, Jupyter notebooks, public datasets, cloud/HPC awareness	1: Get to Know Your Tools 2: Spaceship Interface	Demo / Project Code Review Questions – 1 & 2 Quiz – Projects 1 & 2 BRIDGE-CI Orientation / Resource Awareness
2	Introduction to the Internet of Things (IoT) Projects 3, 4, 5: Electronics, Tinkercad, and Coding NEW Addition BRIDGE-CI Guided Notebook 1A: Open a Jupyter notebook, run prepared cells, load/inspect sensor data, and modify selected settings.	3: Love-O-Meter 4: Color Mixing Lamp 5: Mood Cue	Demo / Project Code Review Questions – 3, 4, & 5 Quiz – Projects 3, 4, & 5 Guided Notebook 1A: basic data exploration
3	IoT Networking Projects 6, 7 & 8: Electronics, Tinkercad, and Coding NEW Addition BRIDGE-CI Guided Notebook 1B: Time-series visualization, rolling averages, thresholds, patterns, and interpretation.	6: Light Theremin 7: Keyboard Instrument 8: Digital Hourglass	Demo / Project Code Review Questions – 6, 7, & 8 Quiz – Projects 6, 7, & 8 Guided Notebook 1B: visualization and interpretation
4	IoT Smart Objects "Things" Projects 9 & 10 Electronics, Tinkercad, and Coding NEW Addition BRIDGE-CI Science Gateway Exploration: Connect small IoT sensor projects to larger shared research tools, datasets, and computing resources.	9: Motorized Pinwheel 10: Zoetrope	Demo / Project Code Review Questions – 9 & 10 Quiz – Projects 9 & 10 Gateway/resource exploration activity

5	IoT Connecting Smart Objects Projects 11 & 12: Electronics, Tinkercad, and Coding IoT Book: Chapter 5 NEW Addition BRIDGE-CI Guided Notebook 2: AI-assisted anomaly and cybersecurity detection; confusion matrix; false positives/false negatives; critical infrastructure consequences.	11: Crystal Ball 12: Knock Lock	Demo / Project Code Review Questions – 11 & 12 Quiz – Projects 11 & 12 Guided Notebook 2: anomaly/cybersecurity interpretation
6	IoT Data and Analytics Final Project Proposal Projects 13, 14, & 15 NEW Addition BRIDGE-CI Final Project Planning: Research question, sensor choice, data-collection plan, CSV structure, and notebook analysis plan.	Work on Final Project	Review Questions & Quiz - Projects 13, 14, 15 BRIDGE-CI Final Project Proposal / Data Plan
7	IoT Privacy and Security NEW Addition BRIDGE-CI Final Project: Collect sensor data, import CSV into Jupyter, analyze and visualize results, interpret findings, and connect the project to cybersecurity or critical infrastructure	Work on Final Project	Final Project BRIDGE-CI Final Project: sensor dataset, notebook, visualizations, and technical report
8	Arduino Certification Exam	Certification Exam	Due by 11:59pm, 3/4 (WED.)